

Whole-genome microarray analyses of *Synechococcus*–*Vibrio* interactions

Vera Tai,¹ Ian T. Paulsen,^{2†} Katherine Phillippy,^{2‡}
D. Aaron Johnson² and Brian Palenik^{1*}

¹Marine Biology Research Division, Scripps Institution of Oceanography, University of California – San Diego, La Jolla, CA 92093-0202, USA.

²J. Craig Venter Institute, Rockville, MD 20850, USA.

Summary

Microbes live in diverse communities yet their physiologies are typically studied in axenic culture. To begin to address this dichotomy, whole-genome microarray analyses were used and revealed that several major metabolic pathways were affected in *Synechococcus* sp. WH8102, a model phototroph, when grown with *Vibrio parahaemolyticus*, a model heterotroph. In co-cultures with *V. parahaemolyticus*, although phosphate was not depleted, *Synechococcus* sp. WH8102 may have experienced phosphate stress since the expression of phosphate acquisition genes increased and alkaline phosphatase activity was higher than in monocultures. Expression of cell wall synthesis genes and the components of a zinc transporter were also upregulated. In contrast, a ferric uptake regulation (Fur) family gene was downregulated as were genes that encode proteins rich in iron or involved in detoxifying oxygen radicals. Nitrogen use may also have been affected in co-cultures as the gene expression changes share similarities with ammonia-grown *Synechococcus*. This study demonstrates the multiple impacts that interspecific microbial interactions can have on the physiology of a major primary producer and the importance of investigating microbial physiology from a community perspective.

Introduction

Axenic cultures are essential for establishing the genetic basis for a microorganism's phenotype and physiology.

Received 17 November, 2008; accepted 4 June, 2009. *For correspondence. E-mail bpalenik@ucsd.edu; Tel. (+1) 858 534 7505; Fax (+1) 858 534 7313. Present addresses: [†]Department of Chemistry and Biomolecular Sciences, Macquarie University, Sydney, NSW 2109, Australia; [‡]National Center for Biotechnology Information, National Library of Medicine, National Institute of Health, Bethesda, MD 20894, USA.

However, under axenic conditions, only intraspecific interactions are occurring. In the environment, microbes exist as part of a community and interspecific interactions could have an important influence on the physiology, growth and ecological niche of each species.

The ecological niches of phytoplankton and heterotrophic bacteria are connected by the 'microbial loop' where bacteria consume and recycle organic carbon and other nutrients originally generated by primary producers. This relationship is not exclusively commensal, however, as bacteria and phytoplankton may also compete for nutrients or provide a resource that the other lacks (Cole, 1982; Grossart and Simon, 2007). For marine systems, most studies have focused on the interactions between eukaryotic phytoplankton, especially bloom-forming species, and heterotrophic bacteria (Haines and Guillard, 1974; Keshtacher-Liebson *et al.*, 1995; Grossart, 1999; Ferrier *et al.*, 2002; Croft *et al.*, 2005) or rely on bulk measurements of bacterial abundance and chlorophyll to infer ecological relationships (Baines and Pace, 1991; Smith *et al.*, 1995; Riemann *et al.*, 2000; Moran *et al.*, 2002). However, the details of the interactions between phytoplankton and bacteria are not well understood.

To investigate the effects of interspecific interactions between phototrophic and heterotrophic bacteria, a co-culture containing the marine cyanobacterium *Synechococcus* sp. strain WH8102 (hereafter referred to as WH8102) and the heterotroph *Vibrio parahaemolyticus* strain LM5312 was established. WH8102 is a representative of the marine *Synechococcus*, cluster 5.1. These are unicellular coccoid cyanobacteria found throughout the world's oceans (Johnson and Sieburth, 1979; Waterbury *et al.*, 1979). The abundance and widespread distribution of marine *Synechococcus* attest to their importance as primary producers in marine ecosystems (Li, 1994). WH8102 was the first marine *Synechococcus* strain to have its genome sequenced (Palenik *et al.*, 2003). The pigment content, absorption spectra, growth irradiance range and other photophysiological parameters also have been characterized for this strain (Six *et al.*, 2004).

Vibrio parahaemolyticus is a heterotrophic marine bacterium known for causing gastroenteritis, but many non-pathogenic strains exist (Joseph *et al.*, 1982; Thompson *et al.*, 2004). The bacteria can survive in the plankton, in sediments, attached to surfaces such as shellfish, or

within a host organism (Joseph *et al.*, 1982). Because *V. parahaemolyticus* can live in a variety of niches, it must be capable of a diverse physiological response. Indeed, the cell type, and thus gene expression, changes whether *V. parahaemolyticus* is living in liquid (swimmer) or on surfaces (swarmer) (McCarter, 1999; Boles and McCarter, 2002). With one of the largest *Vibrio* genomes (Makino *et al.*, 2003), *V. parahaemolyticus* is likely one of the most metabolically versatile.

Vibrio parahaemolyticus are found commonly in marine environments, especially in coastal regions (Colwell *et al.*, 1977; DePaola *et al.*, 1990; Zimmerman *et al.*, 2007). The interaction of *Vibrio* species with *Synechococcus* likely occurs in natural environments as the abundance of *Vibrio* species increases during phytoplankton blooms including blooms of *Synechococcus* (Worden *et al.*, 2006).

Using this phototroph–heterotroph co-culture, physiological changes from the perspective of the cyanobacterium WH8102 when grown with the heterotrophic bacterium *V. parahaemolyticus* were assessed by whole-genome microarray analysis. This is one of the first studies to examine global gene expression changes in bacterial co-cultures.

Results and discussion

We hypothesized that in phototroph–heterotroph co-cultures, the heterotroph would likely be carbon-limited. WH8102 co-cultures were attempted with several species of heterotrophic marine bacteria, but few of the heterotrophs survived serial batch culture with WH8102. WH8102 and *V. parahaemolyticus* were chosen because these are model marine microorganisms with available genome sequences that developed a stable co-culture for further study. As organic carbon was not provided in the media, the success of the WH8102 + *V. parahaemolyticus* co-culture reflects the ability of *V. parahaemolyticus* to survive using the organic carbon produced by WH8102. Likewise, the distribution of microorganisms in marine environments is influenced by the type of organic carbon available (Azam, 1998). As physiological studies of phototrophs are typically conducted with axenic cultures, prior to this study, it was not known whether the phototroph WH8102 would exhibit any physiological or gene expression changes in the presence of a carbon-limited heterotrophic bacterium.

Growth of mono- and co-cultures

WH8102 cell growth in the co-cultures exhibited lag, log and stationary phases, but the length of the phases was slightly different than axenic WH8102 cultures (Fig. 1). The lag phase was longer for WH8102 in co-culture and growth during the exponential phase yielded a lower

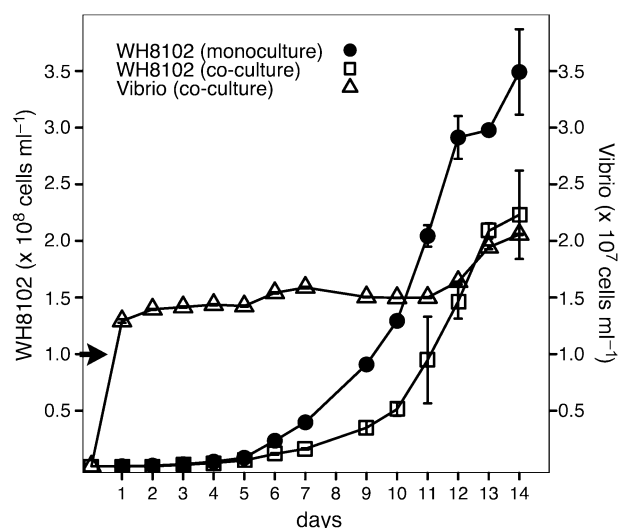


Fig. 1. Sample growth curves of WH8102 in monoculture or co-culture with *V. parahaemolyticus*. WH8102 and *V. parahaemolyticus* cells were counted using flow cytometry. The arrow indicates the approximate concentration in the exponential phase when cells were harvested for RNA extraction. Error bars represent plus or minus the standard deviation.

density of cells. *Vibrio* cells have approximately half the biomass of *Synechococcus* cells (~100 and 200 fg C cell⁻¹, respectively) (Heldal *et al.*, 2003; Lovdal *et al.*, 2008), and after conversion of cell abundance to biomass, the lower yield of WH8102 in the co-cultures was not replaced by *Vibrio* biomass. Based on a curve fitted to the exponential growth phase, WH8102 specific growth rate in the co-culture ($\mu = 0.417 \pm 0.02 \text{ day}^{-1}$, $r^2 = 0.99$) was not significantly different than that of the monoculture ($\mu = 0.416 \pm 0.01 \text{ day}^{-1}$, $r^2 = 0.97$).

In co-cultures, *V. parahaemolyticus* grew exponentially in the first 24 h ($\mu = 0.238 \pm 0.03 \text{ h}^{-1}$, $r^2 = 0.96$), then remained at a static population size (Fig. 1). Later in the time-course, the abundance of *V. parahaemolyticus* cells increased likely due to a rise in extracellular organic carbon release as WH8102 cells entered stationary phase. The addition of glucose or glutamine, but not ammonium, phosphate, iron or zinc, to the WH8102 + *V. parahaemolyticus* co-culture caused a rapid growth of *V. parahaemolyticus*, indicating that *V. parahaemolyticus* growth was limited by organic carbon availability (data not shown).

Gene expression changes in WH8102 when grown with *V. parahaemolyticus*

A statistical analysis of replicated microarray experiments revealed 285 and 295 genes that were significantly upregulated and downregulated, respectively, in WH8102 when grown with *V. parahaemolyticus* (Table 1 and Tables S1 and S2). Genes from several major functional

categories changed in expression level. Transporter and photosynthesis genes comprised the highest number of upregulated genes (Table 1). In addition, genes involved in cell wall biosynthesis, glycosyltransferases and nucleotide biosynthesis were more prominently upregulated rather than downregulated. The most prominent categories of downregulated genes also included transporter and photosynthesis genes. But, in particular, amino acid and cofactor biosynthesis, proteases, post-translational modification and DNA replication, recombination and repair genes were typically downregulated (Table 1). Also downregulated were 4 out of 8 cytochrome b6f genes and 9 out of 16 NADH dehydrogenase genes. In regards to photosynthesis, genes responsible for the synthesis of chlorophyll and phycobilins increased in expression while genes primary related to photosystem II were downregulated. As the co-cultures received adequate amounts of light and the cells were not self-shaded, light limitation does not explain the increase in expression of chlorophyll and phycobilin synthesis genes.

Interesting patterns have emerged from the microarray data, but the reasons for these gene expression changes are complex. As opposed to physiological experiments testing a single factor, the presence of *V. parahaemolyticus* has caused or produced numerous chemical, nutritive or enzymatic factors that were responsible for the

changes in gene expression. By categorizing the changes in the gene expression patterns, the possible causal factors are discussed.

Phosphate stress in WH8102 co-cultures

In cyanobacteria, many of the genes related to phosphate stress are regulated by the same mechanism. For example, low phosphate availability typically activates a transcription factor through a two-component sensor kinase system (Hirani *et al.*, 2001), and the phosphate binding protein, PstS, was found to be specifically expressed when the availability of inorganic phosphate was low (Scanlan *et al.*, 1993). WH8102 is generally expected to respond to phosphate stress similarly (Moore *et al.*, 2005).

Many genes involved in phosphate acquisition and the phosphate stress response were significantly upregulated in WH8102 when grown with *V. parahaemolyticus* and none was downregulated. The phosphate binding protein of an ABC phosphate transporter homologous to PstS (SYNW1018) and a phosphate transcriptional regulator (SYNW1019) were among the most significantly upregulated genes (Table 2). Genes for two additional phosphate transporters and other genes involved in phosphate acquisition also increased in expression, including alka-

Table 1. The number and percentage of genes categorized by function that increased or decreased in expression when WH8102 was grown with *V. parahaemolyticus*.

Function	Increased		Decreased	
	# of genes	% in category	# of genes	% in category
Transporters	16	10.7	18	12.0
Two-component systems	0	0	4	21.1
Proteases	5	11.1	14	31.1
Glycosyltransferases	9	31.1	2	6.9
Cell wall biosynthesis	10	6.8	2	1.4
Amino acid biosynthesis	6	7.5	16	20.0
Cofactor biosynthesis	4	3.4	14	11.8
Nucleotide biosynthesis	6	12.5	0	0
Energy metabolism	10	7.9	6	4.7
Nitrogen metabolism	1	9.1	2	18.2
Other metabolic functions	5	na	9	na
Detoxification, antioxidants	0	na	5	na
Photosynthesis	15	24.6	9	14.8
Ferredoxins	0	0	4	26.7
Electron transport	2	2.5	4	4.9
NADH dehydrogenase	0	0	9	56.3
Transcriptional regulators	4	26.7	3	20.0
Transcription	7	16.3	4	9.3
Translation	8	6.2	6	4.6
Post-translational	1	2.6	14	35.9
DNA replication, recombination, repair	3	4.0	10	13.3
Cell division	1	8.3	6	50.0
Uncategorized	16	na	15	na
Hypothetical	156	14.7	119	11.2
Total	285	11.3	295	11.7

na, not available.

Table 2. Genes related to phosphate acquisition that increased in expression when WH8102 was grown with *V. parahaemolyticus*.

Gene name	Gene ID	Score	Log ₂ fold change
SYNW1019	Possible transcriptional regulator	15.38	1.32
SYNW1018	ABC transporter substrate binding protein phosphate	12.26	2.54
SYNW2390	Putative alkaline phosphatase/5'nucleotidase	11.01	2.64
SYNW0196	Putative alkaline phosphatase	7.76	1.84
SYNW2391	Putative alkaline phosphatase	7.60	2.18
SYNW0762	Conserved hypothetical, possible phytase	6.12	1.46
SYNW1271	Putative phosphate ABC transporter	5.60	0.57
SYNW1286	ABC transporter substrate binding protein component, possibly phosphate	5.33	0.50
SYNW1815	ABC transporter substrate binding protein phosphate	4.84	0.68
SYNW1270	Putative phosphate ABC transporter	4.78	0.31
SYNW1946	PhoH family protein	3.84	0.70

Higher scores reflect stronger statistical significance from the SAM microarray analysis.

line phosphatases (APs) and a possible phytase (Table 2). Based on qRT-PCR, phosphate acquisition genes also showed the greatest increase in transcript level, ranging from 2.73 to 5.34 log₂ fold increases (Table 3).

These increases in the expression of phosphate stress genes occurred despite the availability of inorganic phosphate. Typically, phosphate must be nearly depleted (< 0.05 µM) to induce increased expression and synthesis of phosphate stress genes in cyanobacteria (Ray *et al.*, 1991; Scanlan *et al.*, 1997; Suzuki *et al.*, 2004). The artificial seawater media used here contained 100 µM phosphate and was never completely depleted (Fig. 2A).

Alkaline phosphatase activity in the co-cultures as a measure of phosphate stress

Because AP genes showed increased expression in the co-cultures, it was expected that AP activity would be higher. Total AP activity was consistently higher in the

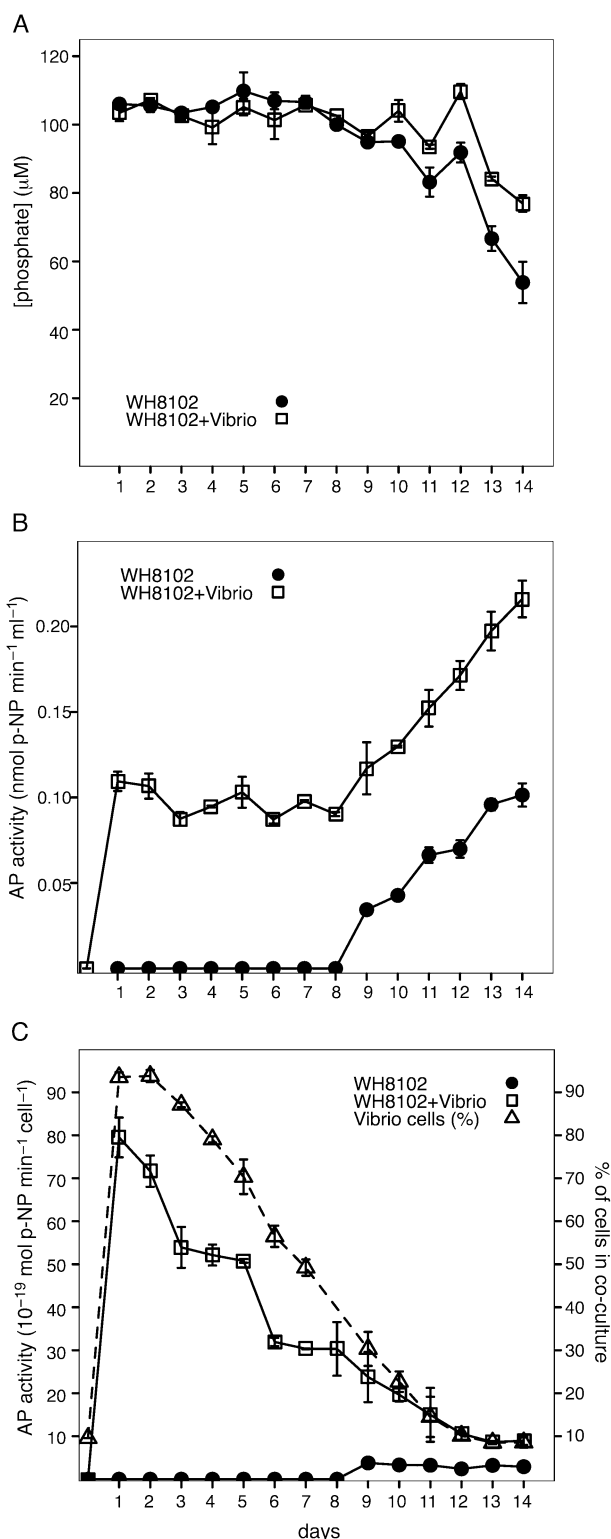
co-culture (Fig. 2B). However, the total AP activity in the co-cultures was likely due to a combination of AP activity produced from both WH8102 and *V. parahaemolyticus* cells.

By calculating AP activity per cell rather than the total AP activity in the co-culture, AP activity per cell decreased as the co-cultures grew (Fig. 2C). The decrease in AP activity per cell matched the decrease in the relative abundance of *V. parahaemolyticus* to WH8102 cells in the co-culture (Fig. 2C). This suggests that *V. parahaemolyticus* initially produced the majority of the AP activity in the co-cultures, but unfortunately, we could not determine the proportion of AP activity ascribed to WH8102 or *V. parahaemolyticus*. As WH8102 cells grew and eventually consisted of 90% of the cells in the co-culture, AP activity was 9.4×10^{-19} mol p-NP min⁻¹ cell⁻¹. Cells were typically harvested for RNA extraction when *V. parahaemolyticus* consisted of ~15% of the cells in the co-culture. At this time point, total AP activity was 15.0×10^{-19} mol p-NP min⁻¹ cell⁻¹.

Table 3. Changes in gene expression using qRT-PCR compared with the microarray analysis.

		RT-qPCR	Microarray	
Gene name	Gene ID	Log ₂ fold change	Score	Log ₂ fold change
Phosphate acquisition				
SYNW1018	ABC transporter substrate binding protein phosphate	3.96 ± 1.64	12.26	2.54
SYNW2390	Putative alkaline phosphatase/5'nucleotidase	5.34 ± 0.10	11.01	2.64
SYNW0196	Putative alkaline phosphatase	2.73 ± 0.16	7.76	1.84
SYNW0762	Conserved hypothetical, possible phytase	3.74 ± 0.21	6.12	1.46
Zinc transport				
SYNW2480	ABC transporter ATP binding component possibly zinc transport	1.00 ± 1.73	13.04	1.95
Secreted proteases				
SYNW2409	Putative haemolysin-type calcium-binding protein; similar to HlyA	0.66 ± 0.77	7.21	1.07
SYNW2304	Possible secreted protease	0.68 ± 1.47	4.62	1.06
Other				
SYNW1139	Hypothetical	1.82 ± 0.25	5.38	1.82
SYNW0512	ATP synthase beta subunit	0.33 ± 0.46	NS	−0.86

In each case the qRT-PCR resulted in the same directional trend as for the microarray analysis. Genes that did not change significantly in expression from the microarray analysis also showed little change using qRT-PCR. For the qRT-PCR results, the mean log₂ fold change ± the standard deviation is reported. Higher absolute scores reflect stronger statistical significance from the SAM analysis. NS indicates that the change in gene expression was not significant.



For *V. parahaemolyticus* phosphate stress-induced AP activity occurs with 500 μM phosphate, while 5000 μM phosphate is replete (McCarter and Silverman, 1987). The media used here contained 100 μM phosphate, so it

Fig. 2. A. Phosphate concentrations in WH8102 monocultures and co-cultures with *V. parahaemolyticus*. B. Total alkaline phosphatase (AP) activity in WH8102 monocultures and co-cultures with *V. parahaemolyticus*. AP activity is reported as nmol of para-nitrophenol (p-NP) produced per minute per ml of culture. C. AP activity per cell in WH8102 monocultures and co-cultures with *V. parahaemolyticus*. AP activity is reported as 10⁻¹⁹ mol of p-NP produced per minute per cell. 'Vibrio cells' is the percentage of total cells that are *V. parahaemolyticus* in the co-culture. Error bars represent plus or minus the standard deviation.

is not surprising that *V. parahaemolyticus* would be contributing to the total AP activity (although growth of *V. parahaemolyticus* in the co-cultures was primarily limited by organic carbon and not phosphate).

In the WH8102 monocultures, AP activity was below the detection limit during the first 8 days of growth (Fig. 2B and C). As the cell number increased, AP activity became detectable at a basal rate of 3.1×10^{-19} mol p-NP min⁻¹ cell⁻¹. This rate is approximately six times lower than AP activity measured from phosphate stressed WH8102 cells (20×10^{-19} mol p-NP min⁻¹ cell⁻¹) (Tetu *et al.*, 2009).

Why are WH8102 cells expressing phosphate stress genes in co-cultures?

Although an increase in AP activity specifically due to WH8102 cells in co-cultures could not be confirmed because of the difficulty partitioning the activity between WH8102 and *V. parahaemolyticus*, we have shown that the expression of SYNW1018, which is homologous to PstS, and other phosphate stress-related genes were induced in *Synechococcus* by the presence of a competing organism with ample phosphate available. Thus, in addition to low phosphate availability, interspecific interactions can also affect phosphate acquisition and stress responses in *Synechococcus*.

The mechanism by which *V. parahaemolyticus* affected expression of WH8102 phosphate-related genes is unknown, but a few possible mechanisms are proposed here. (i) It is possible that *V. parahaemolyticus* secreted a protease that degraded phosphate uptake proteins or phosphate sensors, thus WH8102 may have sensed that phosphate was lacking in the environment. (ii) Pathogenic strains of *V. parahaemolyticus* are known to secrete haemolysins that lyse erythrocytes by forming pores or by phospholipase activity (Zhang and Austin, 2005), and the expression of haemolysins increases under phosphate limitation (McCarter and Silverman, 1987). Their effect on bacterial cells is not known but they might have interfered with phosphate sensing or acquisition. (iii) *V. parahaemolyticus* may have secreted a molecule that affected the regulation of phosphate utilization genes in WH8102. For example, *Vibrio* species are known to excrete quorum sensing molecules, such as acyl homoserine lactones

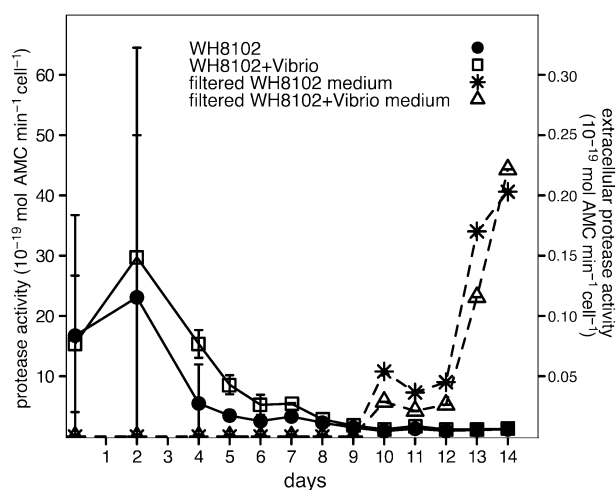


Fig. 3. Total protease activity in WH8102 monocultures and co-cultures with *V. parahaemolyticus*. Protease activity is reported as 10^{-19} mol of AMC produced per minute per cell. Protease activities from filtered media are plotted on the right y-axis. Error bars represent plus or minus the standard deviation.

(Henke and Bassler, 2004). In particular, *V. parahaemolyticus* can secrete molecules that induce gene expression in other *Vibrio* species (Bassler *et al.*, 1997) and this could also occur across more divergent bacteria.

Protease activity in WH8102 co-cultures

As *V. parahaemolyticus* strains are known to secrete lysins and phospholipases, WH8102 may have increased protease activity to protect itself from these antagonistic proteins. Based on the microarray analysis, predicted protease genes both increased and decreased in expression from WH8102 when grown with *V. parahaemolyticus* (Table 1). Those that were secreted or were membrane bound may have interacted with *V. parahaemolyticus*. Two putative secreted proteases (SYNW2304 and SYNW2409) were upregulated as well as one membrane protease (Table 3 and Table S1). Of the downregulated proteases, one was probably secretory (SYNW2176) and three others were membrane-bound (SYNW1476, SYNW1884 and SYNW1909) (Table S2).

Protease activity was slightly higher in co-cultures than in WH8102 monocultures early in the time series. The overall trend, however, was the same (Fig. 3). Protease activity was higher during the first two days of growth, but because protease activity was just above the detection limit during these days, these may not be reliable estimates. Secreted protease activity was measured from filtered media to remove intracellular and membrane bound proteases. Secreted protease activity was not detected until the WH8102 cells reached stationary phase in both mono- and co-cultures (Fig. 3). At this stage, the protease activity observed in the media was likely from

leaking or dying cells. Again, only a slight difference was observed between mono- and co-cultures. The results of these data suggest that a model where a *Vibrio*-produced protease is interacting with *Synechococcus* seems unlikely.

Expression of cell wall modification genes in co-cultured WH8102

A major response of WH8102 to the presence of *V. parahaemolyticus* was the increased expression of cell wall biosynthesis genes (Table 1 and Table S1). In addition, nine glycosyltransferases were also upregulated (Table 1 and Table S1). Glycosyltransferases have many functions in the cell, but they are prominent in the formation of cell walls and exopolysaccharide layers. Ten of these cell wall biosynthesis and glycosyltransferase genes are located together on the genome, transcribed in the same direction, and may function as an operon (Fig. 4). Based on these results, it is likely that WH8102 has modified the composition of its cell wall due to the presence of *V. parahaemolyticus*.

Metal regulation and oxidative stress responses may be linked through a ferric uptake regulation family gene

Genes involved in photosystem II and oxidative stress responses, and genes requiring metal cofactors were often downregulated in WH8102 when grown with *V. parahaemolyticus* (Table 4). These genes are connected physiologically because photosynthesis is a process requiring many metalloenzymes, particularly iron and generates reactive oxygen species (Latifi *et al.*, 2009).

The expression of these genes may be regulated by a ferric uptake regulation (Fur) family gene (SYNW2401). This gene decreased in expression when WH8102 was grown with *V. parahaemolyticus* (Table 4). In *Escherichia coli*, Fur was first recognized as a transcriptional repressor that requires iron as a cofactor and regulates iron acquisition in bacteria by repressing iron transporters and siderophores when iron is sufficient (Bagg and Neilands, 1987; Escolar *et al.*, 1999). Fur family genes also regulate genes related to oxidative stress such as superoxide dismutases (Niederhoffer *et al.*, 1990; Dubrac and Touati, 2000; 2002; Ernst *et al.*, 2005) in addition to acid shock response, chemotaxis, metabolism and virulence (Escolar *et al.*, 1999).

Several genes typically induced by Fur were downregulated in WH8102 when grown with *V. parahaemolyticus* including the cytochrome b6f complex, ferredoxins, superoxide dismutase and other antioxidant enzymes (Table 4). This result may be due to iron limitation in WH8102 when it was co-cultured. A recent study with *E. coli* reports that

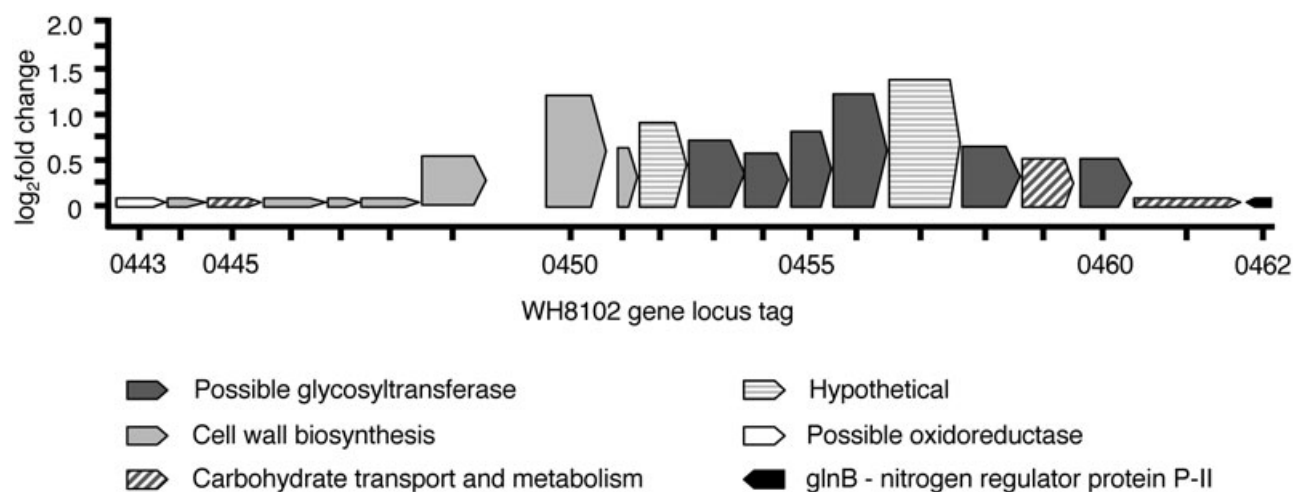


Fig. 4. Potential operon of glycosyltransferases and cell wall biosynthesis genes that was upregulated in WH8102 when grown with *V. parahaemolyticus*. Each arrow represents a gene along the WH8102 chromosome pointing in the direction of transcription. The length of the arrow indicates the size of the gene. The height of the arrow indicates the increase in expression level (expressed as $\log_2$ fold change) compared with WH8102 monocultures. The arrows are coloured by function.

Fur regulates the expression of iron-containing respiratory proteins and that this may be a method for bacteria to mediate iron protein levels based on iron availability (McHugh *et al.*, 2003). A similar homeostatic process may be occurring in WH8102 co-cultures.

Alternatively, SYNW2401 may be zinc-related or a general metal-regulator since zinc-transporters showed increased expression (Table 3 and Table S1). Iron transporters, on the other hand, were not upregulated as would be expected if SYNW2401 was a classical Fe-Fur-type

Table 4. Downregulated genes related to photosynthesis, metal regulation and oxidative stress in WH8102 when grown with *V. parahaemolyticus*.

Gene name	Gene ID	Score	$\log_2$ fold change
Transcriptional regulator			
SYNW2401	Ferric uptake regulator family	-3.57	-0.18
Photosynthesis, downregulated			
SYNW0303	Photosystem II manganese-stabilizing polypeptide	-7.13	-0.63
SYNW0927	Putative photosystem II oxygen-evolving complex PsbP	-7.02	-1.19
SYNW2151	Photosystem II D1 protein form II	-5.74	-2.77
SYNW1982	Photosystem II chlorophyll-binding protein CP47	-4.48	-1.01
SYNW1065	Putative photosystem II reaction centre Psb28 protein	-4.04	-0.41
SYNW1919	Photosystem II D1 protein form II	-3.84	-1.10
SYNW0676	Photosystem II chlorophyll-binding protein CP43	-3.73	-1.37
SYNW2403	Putative high light inducible protein	-3.54	-0.49
SYNW1097	Geranylgeranyl hydrogenase	-3.09	-0.40
Cytochrome b6f complex			
SYNW1967	Apocytochrome b6	-9.69	-1.19
SYNW1841	Cytochrome b6/f complex subunit (Rieske iron-sulfur protein)	-8.70	-1.00
SYNW1842	Apocytochrome f	-8.06	-1.35
SYNW1966	Cytochrome b6-f complex subunit 4 (17 kDa polypeptide)	-3.61	-0.76
Ferredoxins and other iron-rich photosynthetic proteins			
SYNW1277	Ferredoxin	-11.86	-1.12
SYNW0751	Ferredoxin–NADP reductase (FNR)	-6.56	-0.95
SYNW0318	Ferredoxin thioredoxin reductase catalytic beta chain	-6.07	-1.92
SYNW0624	Possible 3Fe-4S ferredoxin	-3.19	-0.60
Detoxification, antioxidants			
SYNW0687	Putative thioredoxin reductase	-7.37	-1.59
SYNW0280	Thioredoxin-like protein TxIA	-7.21	-0.27
SYNW0099	Glutathione peroxidase	-4.23	-1.22
SYNW1280	Alkyl hydroperoxide reductase/Thiol specific antioxidant/Mal allergens family protein	-3.96	-0.31
SYNW1626	putative nickel-containing superoxide dismutase precursor	-3.43	-0.75

Higher absolute scores reflect stronger statistical significance from the SAM analysis.

transcription factor. Often, in addition to an iron-regulated Fur, bacteria have multiple Fur family genes that regulate gene expression for many functions, in particular zinc homeostasis and oxidative stress responses (Hantke, 2001; Andrews *et al.*, 2003). Most cyanobacteria, including all of the fully sequenced marine *Synechococcus* strains, have three Fur family genes, but their exact functions are not known (Hernandez *et al.*, 2004).

Co-cultured WH8102 may use ammonia produced from V. parahaemolyticus

Interestingly, when WH8102 was grown in monoculture with ammonia as opposed to nitrate as the nitrogen source, some similar patterns emerge. In comparison with a microarray analysis of WH8102 cells grown with ammonia (Su *et al.*, 2006), the expression of 75% and 58% of the upregulated and downregulated genes, respectively, changed similarly when WH8102 was grown with *V. parahaemolyticus*. In particular, ammonia-grown WH8102 and WH8102 + *V. parahaemolyticus* co-cultures shared decreased expression of a nitrate transporter (SYNW2463) and urease genes (SYNW2447 and SYNW2449) compared with nitrate-grown WH8102 monocultures (Table S2). However, the global nitrogen regulatory protein, NtcA, did not change in expression level in WH8102 when grown with *V. parahaemolyticus*, whereas it decreased in expression when grown with ammonia as the sole nitrogen source (Su *et al.*, 2006).

These similarities in gene expression patterns could be due to the use of ammonia by WH8102 that was generated by the heterotroph, *V. parahaemolyticus*. Heterotrophic marine bacteria are known to regenerate ammonia particularly if they are carbon-limited (Goldman and Dennett, 1991). Ammonia concentrations measured from the media when the cells were harvested for RNA extraction were approximately 1 μM in both mono- and co-cultures. Although differences in the ammonia concentrations between the mono- and co-cultures were not observed, rapid transport and turnover of the ammonia produced by *V. parahaemolyticus* would explain the lack of an accumulation of ammonia in the co-cultures.

Conclusion

This investigation represents the first study to examine the physiological interactions of a cyanobacterium and a heterotrophic bacterium in co-culture. Even in this simple two-organism community, the presence of *V. parahaemolyticus* caused a multitude of gene expression changes affecting a variety of metabolic pathways in *Synechococcus* sp. WH8102. The mechanisms and reasons for these changes in gene expression could only be surmised and a single causative factor is unlikely, but it is

clear that interspecific interactions have a profound impact on the physiology of a major primary producer. The interactions are undoubtedly more complex in natural communities and we have shown that even at the microbial level, the metabolic functions occurring in an ecosystem are not only dependent on the composition of the community, but also on the interactions among its members.

Experimental procedures

Culture conditions

Axenic WH8102 cultures were maintained in artificial seawater media prepared as described previously (Su *et al.*, 2006) with 9.0 mM NaNO₃. Vitamins (0.2 mg l⁻¹ thiamin-HCl, 1 $\mu\text{g l}^{-1}$ d-biotin and 1 $\mu\text{g l}^{-1}$ B₁₂) were also added. Batch cultures were grown in glass flasks, gently stirred, at 25°C under 30 $\mu\text{Einstein m}^{-2} \text{s}^{-1}$ continuous white light. Under these conditions, WH8102 growth is optimized and does not suffer from light or oxygen limitation. Nutrient analysis of the media was performed by the Oceanographic Data Facility at the Scripps Institution of Oceanography using a Skalar SanPlus continuous-flow AutoAnalyzer.

The WH8102 + *Vibrio parahaemolyticus* co-culture was established by inoculating an axenic culture of WH8102 with a glycerol stock of *V. parahaemolyticus* (strain LM5312, provided by Linda McCarter). The co-culture was maintained as batch cultures by transferring the cells to new media as the WH8102 cells reached late log phase and using the same media and growth conditions as for axenic cultures of WH8102. The addition of organic nutrients was not necessary for growing *V. parahaemolyticus* in the co-culture. By limiting organic resources, the abundance of *V. parahaemolyticus* was kept low relative to WH8102.

For testing the specificity of WH8102 PCR primer pairs, *V. parahaemolyticus* was grown axenically for genomic DNA extractions. The bacteria from the co-culture were streaked onto agar plates made with SN seawater media (Waterbury and Willey, 1988) amended with 0.2% glucose and 0.05% tryptone, and grown at 25°C in the dark to prevent the growth of cyanobacteria for 2–3 days.

Cell counts

Cells were counted using a flow cytometer (FACSort, Becton-Dickinson, Franklin Lakes, NJ). Samples were preserved with 0.25% glutaraldehyde (Sigma, St Louis, MO) for 10 min at room temperature then stored at -80°C. To count cells, the samples were thawed at room temperature, diluted with 0.2 μm filtered media if necessary, and stained with 0.01% SybrGreen I (Invitrogen, Carlsbad, CA) for 15 min in the dark (Lebaron *et al.*, 1998). Just prior to analysis, a known concentration of 0.92 μm , green-fluorescent beads were added (Duke Scientific, Palo Alto, CA). Data was acquired with the FACSort instrument on HI flow for 2 min. WH8102 cells were identified as events with fluorescence in both the FL1 (SybrGreen I) and FL2 (phycoerythrin) channels, whereas *V. parahaemolyticus* cells fluoresced only in FL1.

RNA extraction

From 1 l batch mono- or co-cultures inoculated with 1×10^6 WH8102 cells ml⁻¹, WH8102 cells were grown to mid-log phase (approximately 1×10^8 cells ml⁻¹, see Fig. 1) as described above and harvested by centrifugation at 9000 g for 10 min at room temperature. The cell pellet was resuspended in 15 ml of Trizol reagent (Invitrogen) and incubated at 55°C for 20 min. After adding 3 ml of chloroform, the mixture was shaken vigorously for 30 s and then incubated at room temperature for 10 min. The aqueous phase was recovered after centrifugation at 12 000 g for 15 min at 4°C, mixed with 7.5 ml of isopropanol, and incubated on ice for 12 min. The precipitated RNA was pelleted by centrifugation for 10 min at 12 000 g. The RNA pellet was stored in 75% ethanol at -20°C.

To further purify the RNA, the ethanol was removed from the pellet and the pellet was left to air-dry. The RNA was resuspended in 100 µl of DEPC-treated water. Using the RNeasy Mini kit (Qiagen, Valencia, CA) and following the manufacturer's protocol, the RNA was twice digested with DNase I (Qiagen) and then eluted from the columns. The RNA was quantified by measuring absorbance at 260 nm and stored at -80°C.

From the co-cultures, *V. parahaemolyticus* cells were not separated from the WH8102 cells prior to the RNA extraction, therefore total RNA from the co-cultures was a mix of WH8102 and *V. parahaemolyticus* RNA. Approximately 15% of the cells in the co-culture were *V. parahaemolyticus* at the time of RNA extraction, but the proportion of total RNA that was from *V. parahaemolyticus* was not determined. Equal amounts of total RNA were used to normalize the microarray and qPCR analyses, therefore relatively less WH8102 RNA was assayed from the co-cultures than the monocultures. This means that relative to the monoculture, the magnitude of expression changes in WH8102 from the co-culture is likely greater than observed for upregulated genes and vice versa for downregulated genes. In addition, some truly upregulated genes may have been missed, but on the other hand some downregulated genes may have been falsely identified.

Microarray analysis

A whole-genome microarray for the 2519 genes from *Synechococcus* WH8102 was constructed consisting of a mixed population of PCR amplicons and 70-mer oligonucleotides (for genes smaller than 200 bp) as previously described (Su *et al.*, 2006). PCR products and oligonucleotides were deposited onto Aminosilane-coated UltraGAPS glass slides (Corning, Acton, MA) as described previously (Peterson *et al.*, 2004) using an array spotter (Intelligent Automation Systems, Cambridge, MA).

Four micrograms of total RNA from WH8102 monocultures or WH8102-*Vibrio* co-cultures was labelled with the fluorescent dyes Cy3 or Cy5, respectively, and hybridized to the microarray as described previously (Peterson *et al.*, 2004). RNA from two biological replicates with seven and three technical replicates, respectively, was prepared. For replicates, the dyes were routinely switched to reduce dye-specific biases in signal intensity.

Slides were promptly scanned at a 10 µm resolution using an Axon 4000B scanner with GenePix 4.0 software (Molecular Devices, Sunnyvale, CA). Processing of the TIFF images from hybridized arrays was performed using TIGR-Spotfinder and TIGR-MIDAS (<http://www.tm4.org>). Statistical analysis was performed on log₂-transformed signal ratios of the replicate spots using the Significance Analysis of Microarray (SAM) software (<http://www-stat.stanford.edu/~tibs/SAM/>) (Tusher *et al.*, 2001). Based on the SAM analysis, genes that were significantly upregulated and downregulated had scores greater than 3.11 and lesser than -3.09 respectively. Genes showing significant changes in gene expression were grouped according to functional categories based on the Joint Genome Institute's (JGI) Cluster of Orthologous Groups (COG) categories (http://genome.ornl.gov/microbial/syn_wh/) and gene attributes from the J. Craig Venter Institute – Comprehensive Microbial Resource (<http://cmr.jcvi.org>).

Although the RNA from the co-cultures was a mix of RNA from WH8102 and *V. parahaemolyticus*, *V. parahaemolyticus* RNA was not expected to hybridize to the WH8102 microarray. WH8102 and *V. parahaemolyticus* are divergent bacteria belonging to the *Cyanobacteria* and the *γ-Proteobacteria*, respectively, with only 79% 16S rRNA gene sequence similarity. Even *Synechococcus* strains from different clades do not substantially hybridize to the WH8102 microarray (V. Tai, I. T. Paulsen and B. Palenik, unpubl. data). In addition, we divided the entire *V. parahaemolyticus* genome (strain RIMD 2210633, GenBank accession NC_004603 and NC_004605) into 50 bp fragments and tested the similarity of these sequence fragments to the WH8102 genome (GenBank accession NC_005070). Using the Reference Assembly tool from the CLC Genomics Workbench (version 3.2) set with the default costs (mismatch = 2, insertion = 3, deletion = 3), 100% coverage and 70% similarity, only 72 of the 103 315 *V. parahaemolyticus* sequence fragments aligned with WH8102. Of these, 70 aligned to the rRNA genes and 2 to tRNA genes.

Reverse transcription and quantitative PCR (qRT-PCR)

SuperScript II reverse transcriptase (Invitrogen) was used to synthesize cDNA from 500 ng of RNA from WH8102 monocultures or co-cultures with *V. parahaemolyticus*. The manufacturer's protocol was followed using 100 ng of random hexamer oligonucleotides (Integrative DNA Technologies, Coralville, IA) to prime cDNA synthesis. To test for DNA contamination in the RNA extractions, negative controls with all the reagents except reverse transcriptase were used.

Primers were designed to anneal to the WH8102 ORFs SYNW0196, SYNW0762, SYNW0512, SYNW1018, SYNW1139, SYNW2304, SYNW2390, SYNW2409 and SYNW2480 (Table S3). PCR with genomic DNA from WH8102 and *V. parahaemolyticus*, isolated with the DNeasy kit (Qiagen), was used to test the specificity of the primers. All primer pairs specifically amplified nucleic acids from WH8102, and not from *V. parahaemolyticus*.

Quantitative PCR was performed with a LightCycler instrument (Roche Diagnostics, Indianapolis, IN). Two microlitres of the RT reaction and the LightCycler FastStart DNA Master^{PLUS} SYBR Green I reagent (Roche Diagnostics) was used following the manufacturer's protocol. Tenfold serial dilutions

of genomic WH8102 DNA of known concentration were used to generate a standard curve. The melting curves indicated that RNA from the monocultures and co-cultures produced the same PCR product, indicating that *V. parahaemolyticus* RNA was not amplified. In addition, there was little DNA contamination as negative controls resulted in none to negligible amplification. Amplification efficiencies were at least 84% for all primer pairs.

Alkaline phosphatase and protease activities

One-litre batch mono- and co-cultures inoculated with 1×10^6 WH8102 cell ml⁻¹ were measured daily for cell counts, phosphate concentrations, AP activity and protease activity from 5 ml samples. Culture conditions and cell counts are described above. Phosphate concentration, AP activity and protease activity were measured with unfiltered and 0.2-µm-filtered samples.

Phosphate concentrations were determined by a standard ammonium molybdate assay (Strickland and Parsons, 1972) with the volumes adjusted for measurement with a microplate reader.

Alkaline phosphatase activity was measured by incubating 180 µl of the sample with 20 µl of 135 mM p-nitrophenyl phosphate (Sigma) in 1 M Tris, pH 8.0. Absorbance at 405 nm was read every 30 min for 90 min with a microplate reader. Known concentrations of p-nitrophenol (Sigma) were used to generate a standard curve.

Protease activity was assayed by measuring the production rate of the fluorogenic molecule 7-amino-4-methyl-coumarin (AMC) from the hydrolysis of leucine-AMC (Enzyme Systems Products, Livermore, CA). Samples of 250 µl were incubated with 100 µM of leucine-AMC. Fluorescence (excitation at 370 nm and emission at 440 nm) was measured every 15 min for 90 min with a microplate reader. Dilutions of AMC were used to generate a standard curve.

Enzyme activities are reported as the product produced per minute per total number of cells. For co-cultures, the total number of cells is WH8102 + *V. parahaemolyticus* counts.

Acknowledgements

The authors would like to thank Linda McCarter for providing the *Vibrio parahaemolyticus* culture and Rob Herman for technical assistance. This work was supported by a US Department of Energy grant, DOE DE-FG03-O1ER63148, and by the US Department of Energy's Genomes to Life program (<http://doegenomestolife.org/>) under the project 'Carbon Sequestration in *Synechococcus* sp. From Molecular Machines to Hierarchical Modeling' (<http://www.genomes2life.org>). V.T. also gratefully acknowledges scholarship support from the SIO Graduate Department.

References

- Andrews, S.C., Robinson, A.K., and Rodriguez-Quinones, F. (2003) Bacterial iron homeostasis. *FEMS Microbiol Rev* **27**: 215–237.
- Azam, F. (1998) Microbial control of oceanic carbon flux: the plot thickens. *Science* **280**: 694–696.
- Bagg, A., and Neilands, J.B. (1987) Molecular mechanism of regulation of siderophore-mediated iron assimilation. *Microbiol Rev* **51**: 509–518.
- Baines, S.B., and Pace, M.L. (1991) The production of dissolved organic matter by phytoplankton and its importance to bacteria: patterns across marine and fresh-water systems. *Limnol Oceanogr* **36**: 1078–1090.
- Bassler, B.L., Greenberg, E.P., and Stevens, A.M. (1997) Cross-species induction of luminescence in the quorum-sensing bacterium *Vibrio harveyi*. *J Bacteriol* **179**: 4043–4045.
- Boles, B.R., and McCarter, L.L. (2002) *Vibrio parahaemolyticus* scrABC, a novel operon affecting swarming and capsular polysaccharide regulation. *J Bacteriol* **184**: 5946–5954.
- Cole, J.J. (1982) Interactions between bacteria and algae in aquatic ecosystems. *Annu Rev Ecol Syst* **13**: 291–314.
- Colwell, R.R., Kaper, J., and Joseph, W. (1977) *Vibrio cholerae*, *Vibrio parahaemolyticus*, and other *Vibrios* – occurrence and distribution in Chesapeake Bay. *Science* **198**: 394–396.
- Croft, M.T., Lawrence, A.D., Raux-Deery, E., Warren, M.J., and Smith, A.G. (2005) Algae acquire vitamin B-12 through a symbiotic relationship with bacteria. *Nature* **438**: 90–93.
- DePaola, A., Hopkins, L.H., Peeler, J.T., Wentz, B., and McPhearson, R.M. (1990) Incidence of *Vibrio parahaemolyticus* in U.S. coastal waters and oysters. *Appl Environ Microbiol* **56**: 2299–2302.
- Dubrac, S., and Touati, D. (2000) Fur positive regulation of iron superoxide dismutase in *Escherichia coli*: functional analysis of the *sodB* promoter. *J Bacteriol* **182**: 3802–3808.
- Dubrac, S., and Touati, D. (2002) Fur-mediated transcriptional and post-transcriptional regulation of FeSOD expression in *Escherichia coli*. *Microbiology-SGM* **148**: 147–156.
- Ernst, F.D., Homuth, G., Stoof, J., Mader, U., Waidner, B., Kuipers, E.J., et al. (2005) Iron-responsive regulation of the *Helicobacter pylori* iron-cofactored superoxide dismutase *SodB* is mediated by *fur*. *J Bacteriol* **187**: 3687–3692.
- Escolar, L., Perez-Martin, J., and De Lorenzo, V. (1999) Opening the iron box: transcriptional metalloregulation by the *fur* protein. *J Bacteriol* **181**: 6223–6229.
- Ferrier, M., Martin, J.L., and Rooney-Varga, J.N. (2002) Stimulation of *Alexandrium fundyense* growth by bacterial assemblages from the Bay of Fundy. *J Appl Microbiol* **92**: 706–716.
- Goldman, J.C., and Dennett, M.R. (1991) Ammonium regeneration and carbon utilization by marine bacteria grown on mixed substrates. *Mar Biol* **109**: 369–378.
- Grossart, H.P. (1999) Interactions between marine bacteria and axenic diatoms (*Cylindrotheca fusiformis*, *Nitzschia laevis*, and *Thalassiosira weissflogii*) incubated under various conditions in the lab. *Aquat Microb Ecol* **19**: 1–11.
- Grossart, H.P., and Simon, M. (2007) Interactions of planktonic algae and bacteria: effects on algal growth and organic matter dynamics. *Aquat Microb Ecol* **47**: 163–176.
- Haines, K.C., and Guillard, R.R. (1974) Growth of vitamin B12-requiring marine diatoms in mixed laboratory cultures with vitamin B12-producing marine bacteria. *J Phycol* **10**: 245–252.

- Hantke, K. (2001) Iron and metal regulation in bacteria. *Curr Opin Microbiol* **4**: 172–177.
- Heldal, M., Scanlan, D.J., Norland, S., Thingstad, F., and Mann, N.H. (2003) Elemental composition of single cells of various strains of marine *Prochlorococcus* and *Synechococcus* using X-ray microanalysis. *Limnol Oceanogr* **48**: 1732–1743.
- Henke, J.M., and Bassler, B.L. (2004) Bacterial social engagements. *Trends Cell Biol* **14**: 648–656.
- Hernandez, J.A., Lopez-Gomollon, S., Bes, M.T., Fillat, M.F., and Peleato, M.L. (2004) Three fur homologues from *Anabaena* sp. PCC7120: exploring reciprocal protein-promoter recognition. *FEMS Microbiol Lett* **236**: 275–282.
- Hirani, T.A., Suzuki, I., Murata, N., Hayashi, H., and Eaton-Rye, J.J. (2001) Characterization of a two-component signal transduction system involved in the induction of alkaline phosphatase under phosphate-limiting conditions in *Synechocystis* sp. PCC 6803. *Plant Mol Biol* **45**: 133–144.
- Johnson, P.W., and Sieburth, J.M. (1979) Chroococcoid cyanobacteria in the sea: a ubiquitous and diverse phototropic biomass. *Limnol Oceanogr* **24**: 928–935.
- Joseph, S.W., Colwell, R.R., and Kaper, J.B. (1982) *Vibrio parahaemolyticus* and related halophilic vibrios. *CRC Crit Rev Microbiol* **10**: 77–124.
- Keshtacher-Liebson, E., Hadar, Y., and Chen, Y. (1995) Oligotrophic bacteria enhance algal growth under iron-deficient conditions. *Appl Environ Microbiol* **61**: 2439–2441.
- Latifi, A., Ruiz, M., and Zhang, C.C. (2009) Oxidative stress in cyanobacteria. *FEMS Microbiol Rev* **33**: 258–278.
- Lebaron, P., Parthuisot, N., and Catala, P. (1998) Comparison of blue nucleic acid dyes for flow cytometric enumeration of bacteria in aquatic systems. *Appl Environ Microbiol* **64**: 1725–1730.
- Li, W.K.W. (1994) Primary production of prochlorophytes, cyanobacteria, and eukaryotic ultraphytoplankton: measurements from flow cytometric sorting. *Limnol Oceanogr* **39**: 169–175.
- Lovdal, T., Skjoldal, E.F., Heldal, M., Norland, S., and Thingstad, T.F. (2008) Changes in morphology and elemental composition of *Vibrio splendidus* along a gradient from carbon-limited to phosphate-limited growth. *Microb Ecol* **55**: 152–161.
- Makino, K., Oshima, K., Kurokawa, K., Yokoyama, K., Uda, T., Tagomori, K., et al. (2003) Genome sequence of *Vibrio parahaemolyticus*: a pathogenic mechanism distinct from that of *V. cholerae*. *Lancet* **361**: 743–749.
- McCarter, L.L. (1999) The multiple identities of *Vibrio parahaemolyticus*. *J Mol Microbiol Biotechnol* **1**: 1–7.
- McCarter, L.L., and Silverman, M. (1987) Phosphate regulation of gene expression in *Vibrio parahaemolyticus*. *J Bacteriol* **169**: 3441–3449.
- McHugh, J.P., Rodriguez-Quinones, F., Abdul-Tehrani, H., Svistunenko, D.A., Poole, R.K., Cooper, C.E., and Andrews, S.C. (2003) Global iron-dependent gene regulation in *Escherichia coli* – A new mechanism for iron homeostasis. *J Biol Chem* **278**: 29478–29486.
- Moore, L.R., Ostrowski, M., Scanlan, D.J., Feren, K., and Sweetsir, T. (2005) Ecotypic variation in phosphorus acquisition mechanisms within marine picocyanobacteria. *Aquat Microb Ecol* **39**: 257–269.
- Moran, X.A.G., Estrada, M., Gasol, J.M., and Pedros-Alio, C. (2002) Dissolved primary production and the strength of phytoplankton bacterioplankton coupling in contrasting marine regions. *Microb Ecol* **44**: 217–223.
- Niederhoffer, E.C., Naranjo, C.M., Bradley, K.L., and Fee, J.A. (1990) Control of *Escherichia coli* superoxide dismutase (SodA and SodB) genes by the ferric uptake regulation (Fur) locus. *J Bacteriol* **172**: 1930–1938.
- Palenik, B., Brahamsha, B., Larimer, F.W., Land, M., Hauser, L., Chain, P., et al. (2003) The genome of a motile marine *Synechococcus*. *Nature* **424**: 1037–1042.
- Peterson, S.N., Sung, C.K., Cline, R., Desai, B.V., Snesrud, E.C., Luo, P., et al. (2004) Identification of competence pheromone responsive genes in *Streptococcus pneumoniae* by use of DNA microarrays. *Mol Microbiol* **51**: 1051–1070.
- Ray, J.M., Bhaya, D., Block, M.A., and Grossman, A.R. (1991) Isolation, transcription, and inactivation of the gene for an atypical alkaline-phosphatase of *Synechococcus* sp. strain PCC 7942. *J Bacteriol* **173**: 4297–4309.
- Riemann, L., Steward, G.F., and Azam, F. (2000) Dynamics of bacterial community composition and activity during a mesocosm diatom bloom. *Appl Environ Microbiol* **66**: 578–587.
- Scanlan, D.J., Mann, N.H., and Carr, N.G. (1993) The response of the picoplanktonic marine cyanobacterium *Synechococcus* species WH7803 to phosphate starvation involves a protein homologous to the periplasmic phosphate-binding protein of *Escherichia coli*. *Mol Microbiol* **10**: 181–191.
- Scanlan, D.J., Silman, N.J., Donald, K.M., Wilson, W.H., Carr, N.G., Joint, I., and Mann, N.H. (1997) An immunological approach to detect phosphate stress in populations and single cells of photosynthetic picoplankton. *Appl Environ Microbiol* **63**: 2411–2420.
- Six, C., Thomas, J.C., Brahamsha, B., Lemoine, Y., and Partensky, F. (2004) Photophysiology of the marine cyanobacterium *Synechococcus* sp. WH8102, a new model organism. *Aquat Microb Ecol* **35**: 17–29.
- Smith, D.C., Steward, G.F., Long, R.A., and Azam, F. (1995) Bacterial mediation of carbon fluxes during a diatom bloom in a mesocosm. *Deep Sea Res Part II* **42**: 75–97.
- Strickland, J.D., H., and Parsons, T.R. (1972) *A Practical Handbook of Seawater Analysis*. Ottawa, Canada: Fisheries Research Board of Canada.
- Su, Z.C., Mao, F.L., Dam, P., Wu, H.W., Olman, V., Paulsen, I.T., et al. (2006) Computational inference and experimental validation of the nitrogen assimilation regulatory network in cyanobacterium *Synechococcus* sp. WH 8102. *Nucleic Acids Res* **34**: 1050–1065.
- Suzuki, S., Ferjani, A., Suzuki, I., and Murata, N. (2004) The SphS-SphR two component system is the exclusive sensor for the induction of gene expression in response to phosphate limitation in *Synechocystis*. *J Biol Chem* **279**: 13234–13240.
- Tetu, S.G., Brahamsha, B., Johnson, D.A., Tai, V., Phillippy, K., Palenik, B., and Paulsen, I.T. (2009) Microarray analysis of phosphate regulation in the marine cyanobacterium *Synechococcus* sp. WH8102. *ISME J* **3**: 835–849.

- Thompson, F.L., Iida, T., and Swings, J. (2004) Biodiversity of vibrios. *Microbiol Mol Biol Rev* **68**: 403–431.
- Tusher, V.G., Tibshirani, R., and Chu, G. (2001) Significance analysis of microarrays applied to the ionizing radiation response. *Proc Natl Acad Sci USA* **98**: 5116–5121.
- Waterbury, J.B., Watson, S.W., Guillard, R.R.L., and Brand, L.E. (1979) Widespread occurrence of a unicellular, marine, planktonic, cyanobacterium. *Nature* **277**: 293–294.
- Waterbury, J.B., and Willey, J.M. (1988) Isolation and growth of marine planktonic cyanobacteria. *Methods Enzymol* **167**: 100–105.
- Worden, A.Z., Seidel, M., Smriga, S., Wick, A., Malfatti, F., Bartlett, D., and Azam, F. (2006) Trophic regulation of *Vibrio cholerae* in coastal marine waters. *Environ Microbiol* **8**: 21–29.
- Zhang, X.H., and Austin, B. (2005) Haemolysins in *Vibrio* species. *J Appl Microbiol* **98**: 1011–1019.
- Zimmerman, A.M., DePaola, A., Bowers, J.C., Krantz, J.A., Nordstrom, J.L., Johnson, C.N., and Grimes, D.J. (2007) Variability of total and pathogenic *Vibrio parahaemolyticus* densities in northern Gulf of Mexico water and oysters. *Appl Environ Microbiol* **73**: 7589–7596.

Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. List of significantly upregulated genes in WH8102 when grown with *V. parahaemolyticus* compared with WH8102 monocultures. Higher scores from the SAM analysis reflect stronger statistical significance. Genes are grouped based on their probable function.

Table S2. List of significantly downregulated genes in WH8102 when grown with *V. parahaemolyticus* compared with WH8102 monocultures. Higher absolute scores from the SAM analysis reflect stronger statistical significance. Genes are grouped based on their probable function.

Table S3. Primers used for qRT-PCR and expected sizes of the PCR products. The primers pairs were specific to WH8102 and did not amplify nucleic acids from *V. parahaemolyticus*.

Please note: Wiley-Blackwell are not responsible for the content or functionality of any supporting materials supplied by the authors. Any queries (other than missing material) should be directed to the corresponding author for the article.